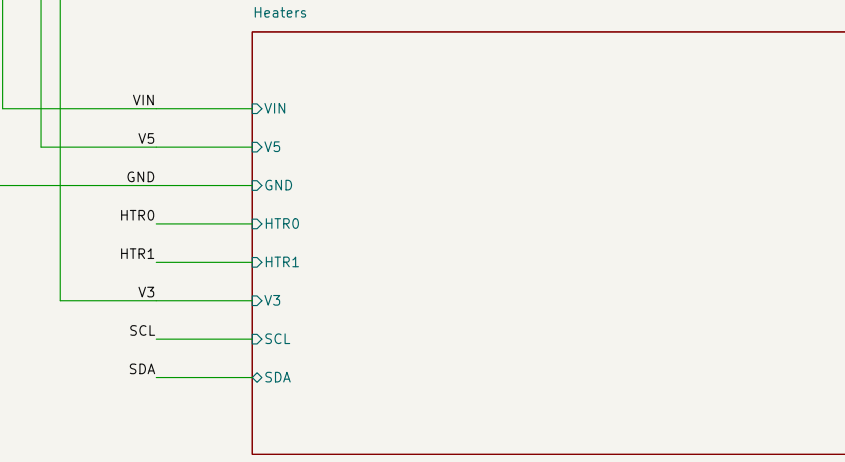


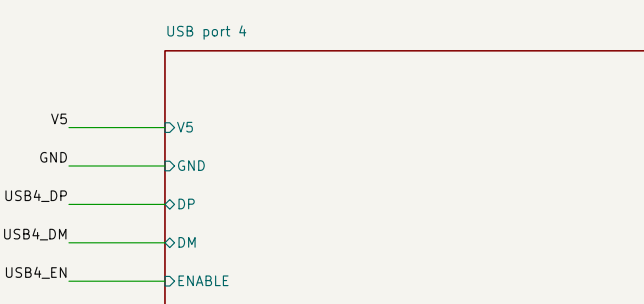
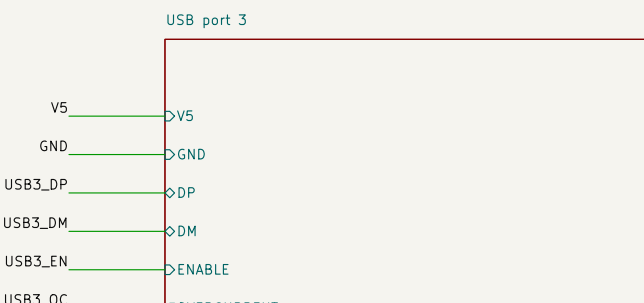
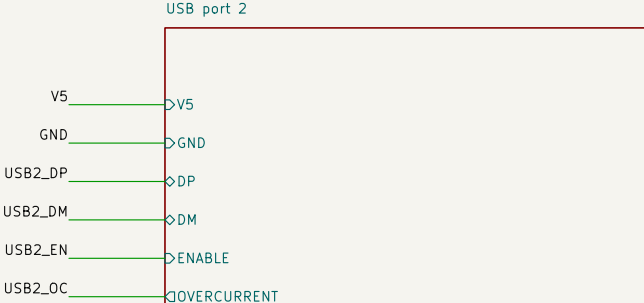
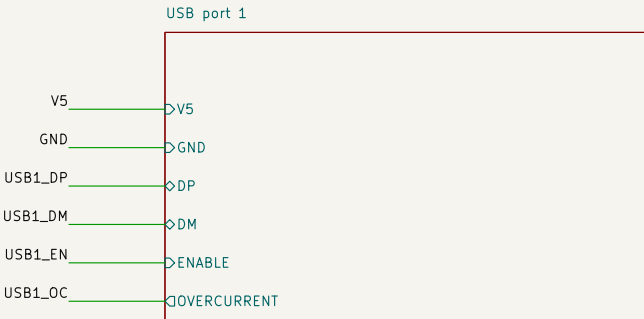
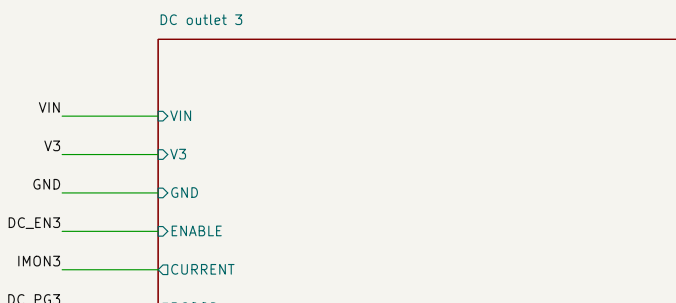
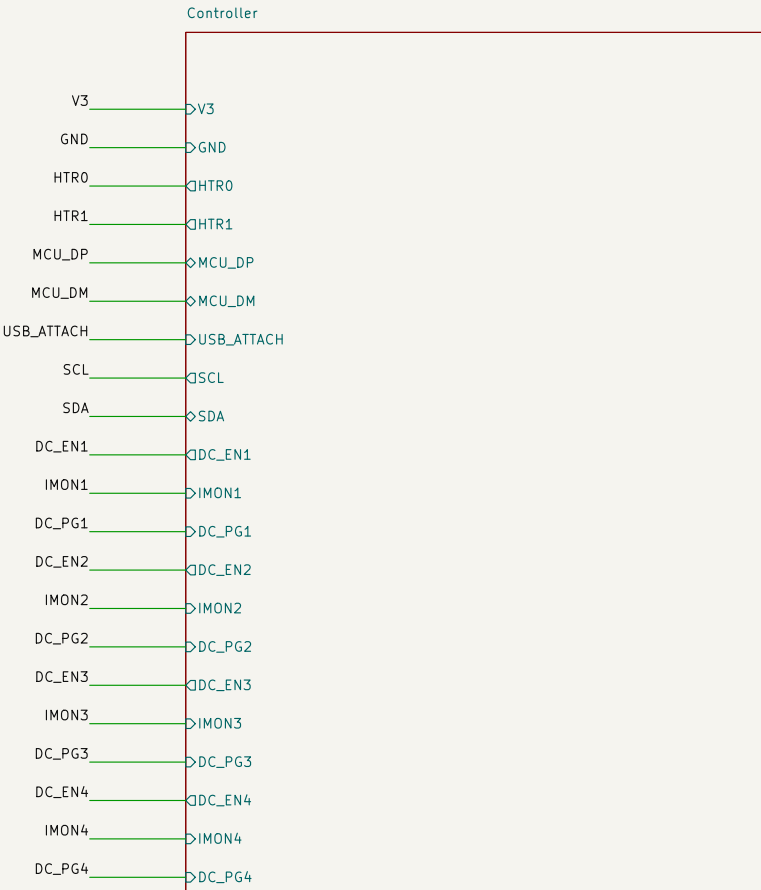
PULSARDEW HUB | 2 heaters + 4 monitored DC outlets + 4 USB 2.0 ports

PROTOTYPE FLOORPLAN — routing, full-load thermal validation and new firmware are still required.

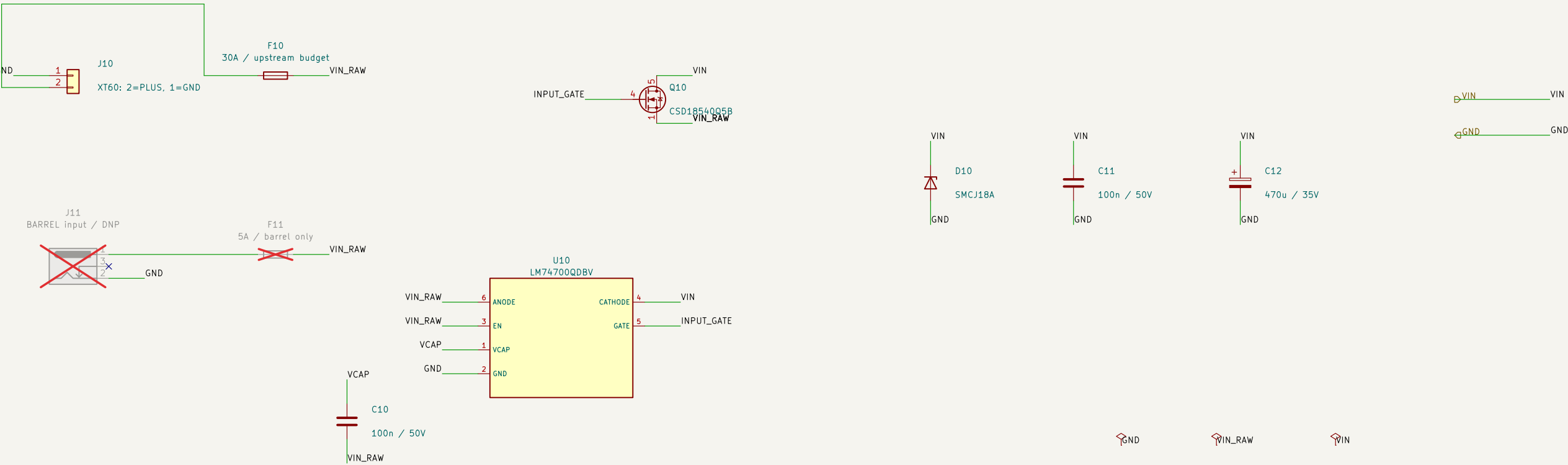


RATINGS TO VALIDATE
12–18 V nominal input
4 x 5 A DC outlet targets
2 x 5 A heater targets
25 A shared XT60 budget
3.5 A continuous barrel budget
USB: 500 mA per port
No PWM on DC outlets

Same ground throughout.
High-current and USB return paths
are separated by placement.
Do not backfeed DC outputs.

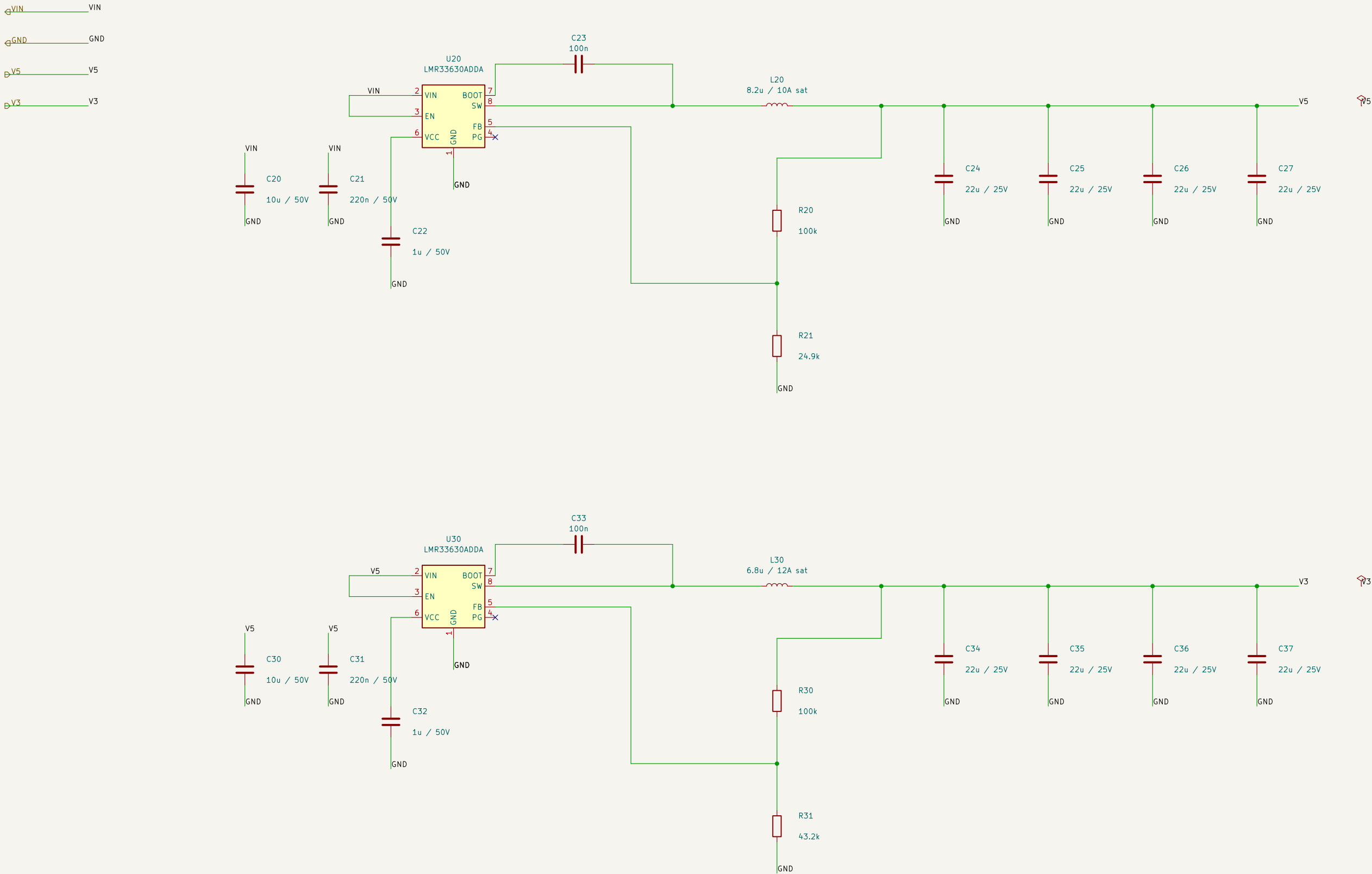


POWER INPUT | 12–18 V nominal | XT60 or barrel assembly option



FIT J10 + F10 OR J11 + F11. Do not populate both input connector paths.
XT60: provisional 25 A shared continuous budget; fuse, copper and temperature still require validation.
Barrel: 5 A absolute TOTAL ceiling; 3.5 A provisional continuous budget after fuse derating. Use a current-limited supply.
LM74700 + Q10 provides reverse-polarity / reverse-current protection. It is not an overvoltage disconnect.
Never apply more than 18 V nominal. TVS pulse capability is not a sustained overvoltage rating.

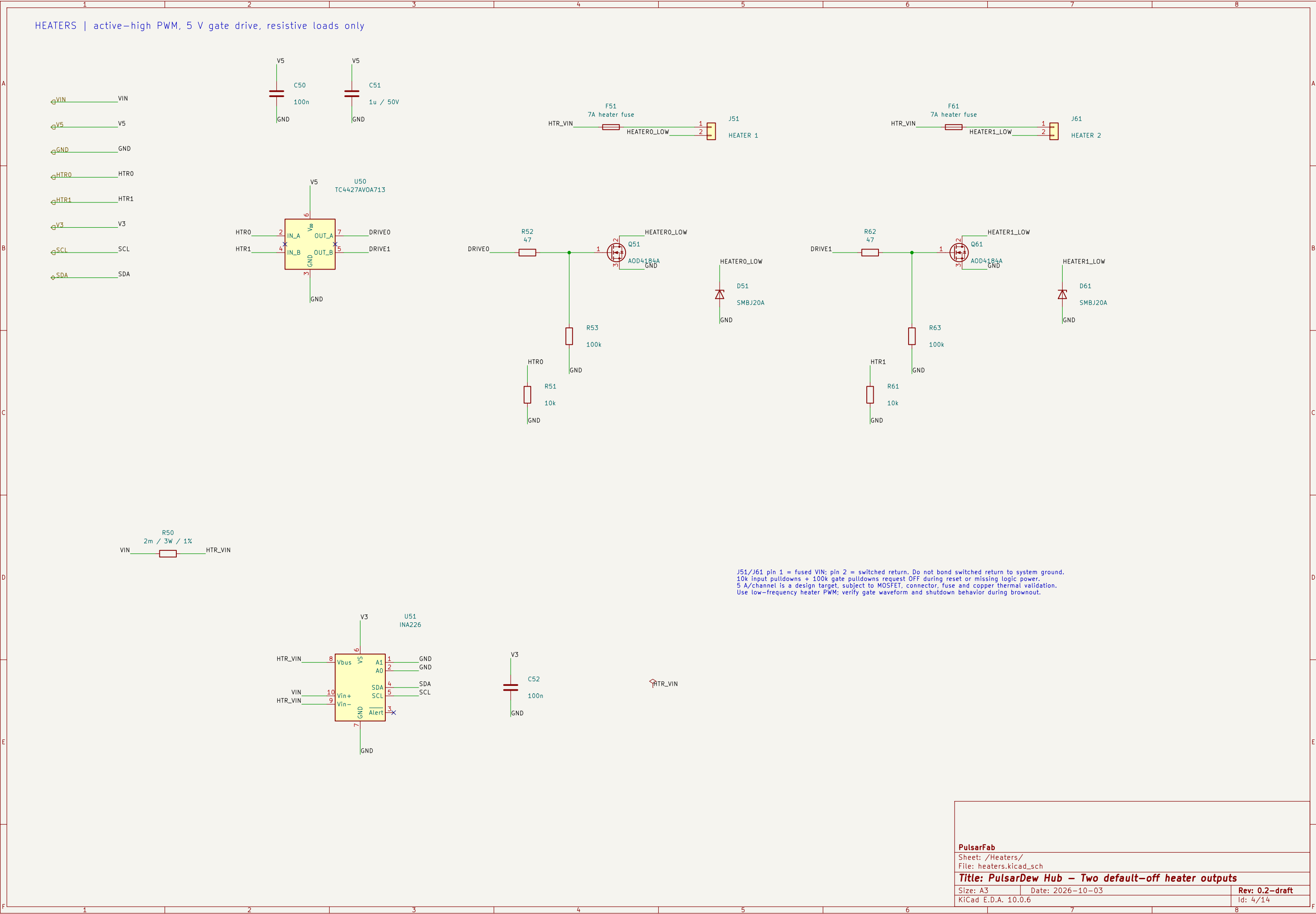
DC SUPPLIES | LMR33630ADDA, 400 kHz | 5 V / 3 A + 3.3 V logic



Follow TI SNVSN3F Table 9–2. 5 V uses nearest preferred 8.2 uH to the 8 uH example.
Place ceramic input/boot/VCC capacitors at the IC; keep SW copper compact and FB away from SW.
USB budget: 4 x 500 mA plus about 0.4 A for logic at full activity. Verify inductor saturation, MLCC bias, transient response and regulator temperature before routing release.

PulsarFab		
Sheet: /DC supplies/ File: supplies.kicad_sch		
Title: PulsarDew Hub – 5 V USB and 3.3 V logic supplies		
Size: A3	Date: 2026–10–03	Rev: 0.2–draft
KiCad E.D.A. 10.0.6		Id: 3/14

HEATERS | active-high PWM, 5 V gate drive, resistive loads only



12345678

CONTROLLER | 2 heater PWM + 4 DC enables / ADC current readings / faults

A

B

C

D

E

F

V3 C1 100n GND

V3 C2 4.7u / 16V GND

V3 C3 100n GND

NRST

DC_EN4

HTR0

HTR1

IMON1

IMON2

IMON3

IMON4

SWDIO

MCU_DM

MCU_DP

USB_ATTACH

SCL

SDA

U1 STM32G0B1KBU6

VDD

PA0

PA1

PA2

PA3

PA4

PA5

PA6

PA7

PA8

PA13/SWDIO

PA11

PA12

PA14/SWCLK/BOOT0

YSS

EP

GND

DC_EN1

DC_EN2

DC_EN3

USB_ATTACH

DC_PG4

DC_PG1

DC_PG2

DC_PG3

SCL

SDA

SWCLK

V3

R1 4.7k SCL

R2 4.7k SDA

U2 SHT40-AD1B-R2

SCL

SDA

VSS

VDD

GND

V3 C4 100n GND

J1

V3 1

SWDIO 2

SWCLK 3

NRST 4

GND 5

SWD

V3

GND

DC_EN1

DC_EN2

DC_EN3

DC_EN4

IMON1

IMON2

IMON3

IMON4

DC_PG1

DC_PG2

DC_PG3

DC_PG4

V3

GND

PA0/PA1: active-high TIM2 heater drive. PB0/PB1/PB2/PB9: static DC GPIO only (never PWM).
PA4..PA7: ADC IMON1..4. PB3/PB4/PB5/PC6: active-low DC faults.
PA9: internal hub port 1 enable (3.3 V). Attach USB only while high; disable UCPD dead-battery pull-down.
SHT40 at 0x44; keep physically away from power components. New firmware pin map is required.

PulsarFab

Sheet: /Controller/
File: controller.kicad_sch

Title: PulsarDew Hub – STM32 controller and ambient sensor

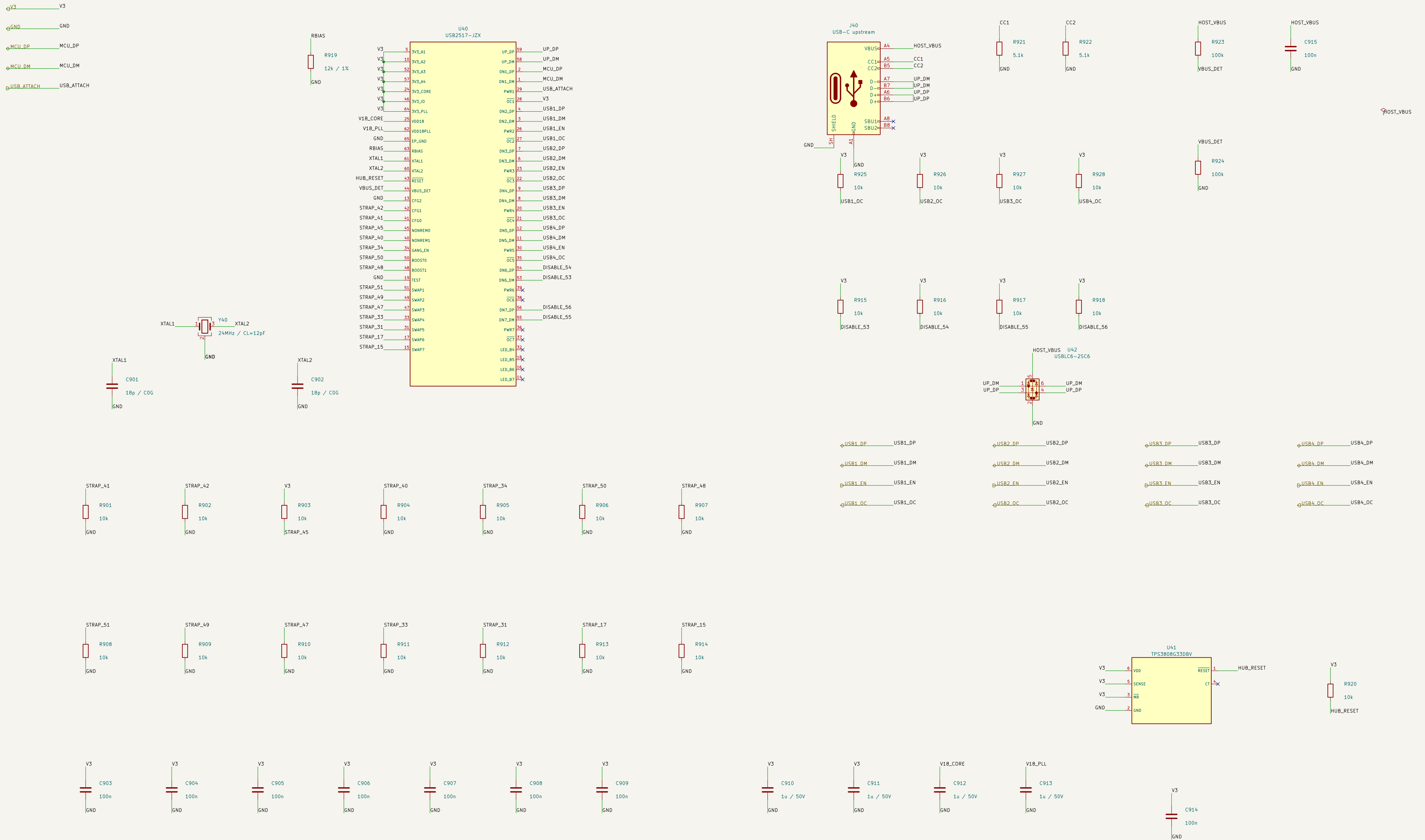
Size: A3 Date: 2026-10-03 Rev: 0.2-draft

KiCad E.D.A. 10.0.6 Id: 5/14

12345678

Sheet: /Controller/ File: controller.kicad_sch		
Title: PulsarDew Hub – STM32 controller and ambient sensor		
Size: A3	Date: 2026–10–03	Rev: 0.2–draft
KiCad E.D.A. 10.0.6		Id: 5/14

USB HUB | USB2517 | internal port 1 + external ports 2-5; ports 6/7 disabled



Straps: CFG=000 self-powered; NONREM=01 makes port 1 permanent; individual power/OC; normal polarity. Unused ports 6/7; both D+ and D- pulled to 3V3 through 10k; PWR outputs unused, OC inputs unused. VDD18 and VDD18PLL are separate INTERNAL regulator outputs; capacitors only; do not connect them together. Supervisor: 3.07 V threshold, 20 ms delay. Crystal caps start at 18 pf; verify actual board parasitics/drive level. Keep an uninterrupted ground plane under every USB pair. Verify 90-ohm geometry with the ordered stackup.

SWITCHED DC OUTLET | 12–18 V / 5 A service | hardware voltage inhibit / latched breaker

VIN VIN
V3 V3
GND GND
ENABLE ENABLE
CURRENT CURRENT
PGOOD PGOOD

VIN VIN
D102 SMBJ18A
GND GND
C101 100n / 50V
GND GND

U101 TPS2598270NRGE
VIN 1 IN OUT 17 OUT
VIN 2 IN OUT 18 OUT
VIN 3 IN OUT 19 OUT
VIN 16 IN OUT 20 OUT
VIN 25 EP_IN OUT 21 OUT
EN_SW 6 EN_UVLO OUT 22 OUT
GND 4 GND OUT 23 OUT
GND 5 GND OUT 24 OUT
GND 14 GND PG PGOOD
GND 26 EP_GND IMON IMON_RAW
IMON 8 ILIM
dVdT 15 DVDT
ITIMER 7 ITIMER
RETRY_DLY 10 RETRY_DLY
NRETRY 11 NRETRY
LDSTRT 12 LDSTRT

OUT OUT
C102 100n / 50V
GND GND

J101 5A DC / center +
OUT OUT
GND GND

V3 V3
R106 10k
PGOOD PGOOD

OUT OUT
D101 SS54
GND GND

VIN VIN
R101 220k UV_SENSE
UV_SENSE UV_SENSE
R102 10k GND
GND GND
R103 470k OV_SENSE
OV_SENSE OV_SENSE
R104 10k GND
GND GND
U102 TPS3700DDC
V3 5 VDD OUTA 1 EN_SW
UV_SENSE 3 INA+ OUTB 6 EN_SW
OV_SENSE 4 INB-
GND 2 GND

V3 V3
C105 100n
GND GND

ENABLE ENABLE
R110 10k EN_SW
EN_SW EN_SW
DVDT DVDT
C103 4.7n / 50V
GND GND
ILIM ILIM
R107 249 / 1%
GND GND
EN_SW EN_SW
R105 100k
GND GND

IMON_RAW IMON_RAW
R109 10k CURRENT
CURRENT CURRENT
C104 10n
GND GND

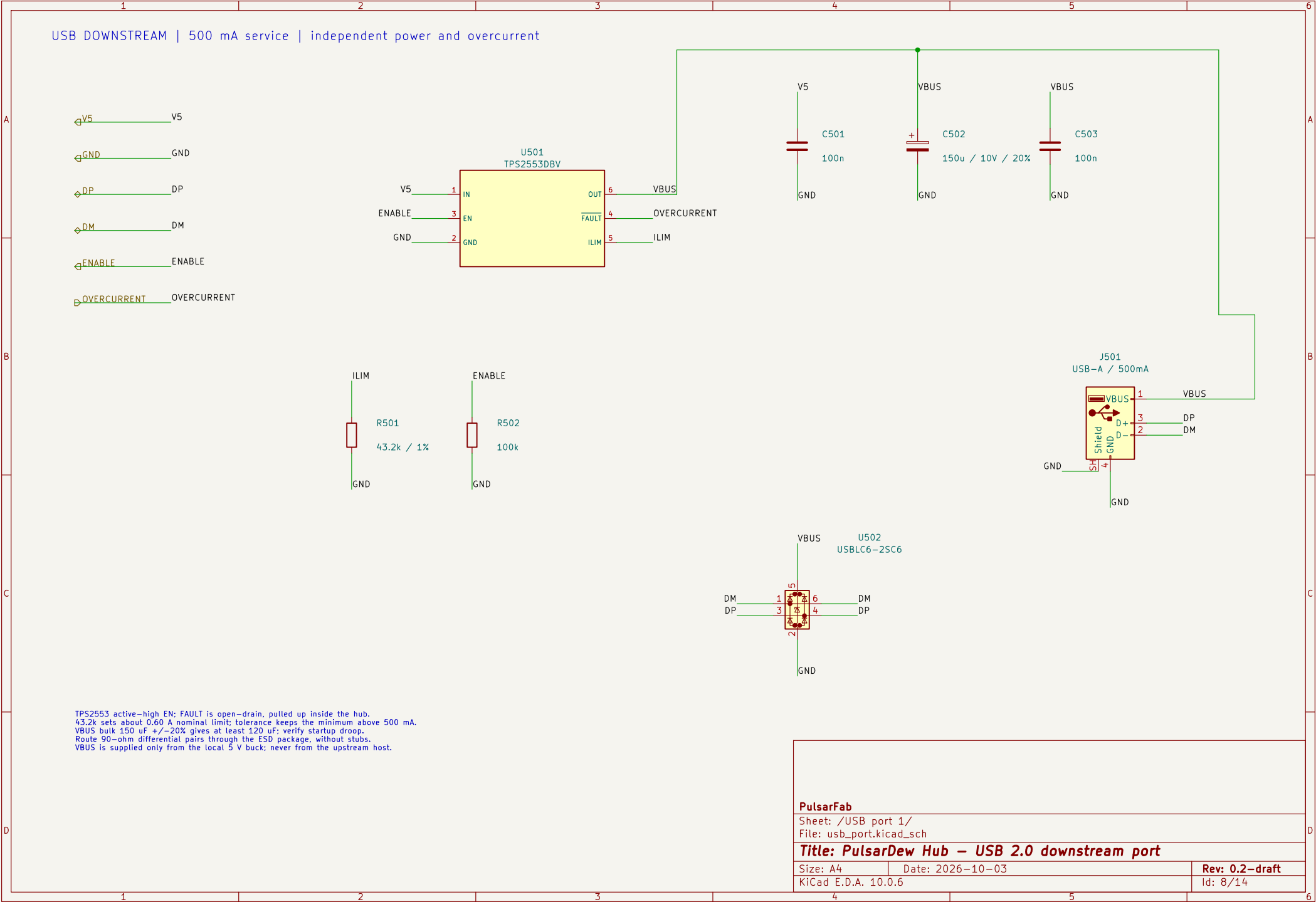
IMON_RAW IMON_RAW
R108 820 / 1%
GND GND

TPS2598270: 249R gives 5.97 A nominal breaker threshold; 5 A service target needs tolerance/bench qualification.
ITIMER open: fastest overload response. Fast short trip ~2.1x setpoint. RETRY_DLY grounded: latch off.
IMON: 246 uA/A x 820R = 0.20172 V/A (1.009 V at 5 A); 10k / 10n ADC filter. Calibrate each outlet.
TPS3700 independently inhibits EN below ~9.2 V or above ~19.2 V; window recovery can re-enable an asserted GPIO.
PGOOD is HIGH when ready; LOW also means OFF/startup/inhibit, not exclusively a latched current fault.
EP25 = VIN, EP26 = GND. -15 C minimum junction rating accepted. No reverse blocking; do not backfeed OUT.
SMBJ18A input TVS + SS54 output clamp need short loops and hot-plug/inductive-load bench validation.

PulsarFab
Sheet: /DC outlet 1/
File: dc_outlet.kicad_sch
Title: PulsarDew Hub – 5 A switched DC outlet
Size: A3 Date: 2026–10–03 Rev: 0.2–draft
KiCad E.D.A. 10.0.6 Id: 7/14

TPS2598270: 249R gives 5.97 A nominal breaker threshold; 5 A service target needs tolerance/bench qualification. ITIMER open: fastest overload response. Fast short trip ~2.1x setpoint. RETRY_DLY grounded: latch off. IMON: 246 uA/A x 820R = 0.20172 V/A (1.009 V at 5 A); 10k / 10n ADC divider. Calibrate each outlet. TPS37700 independently inhibits EN below -9.2 V or above -19.2 V: window recovery can re-enable an asserted GPIO. PGOOD is HIGH when ready; LOW also means OFF/startup/inhibit, not exclusively a latched current fault. EPB2 = VIN, EP26 = GND. -15 C minimum junction rating accepted. No reverse blocking; do not backfeed OUT. SMJ18A input TVS + S554 output clamp need short loops and hot+plug/inductive-load bench validation.

USB DOWNSTREAM | 500 mA service | independent power and overcurrent



PulsarFab

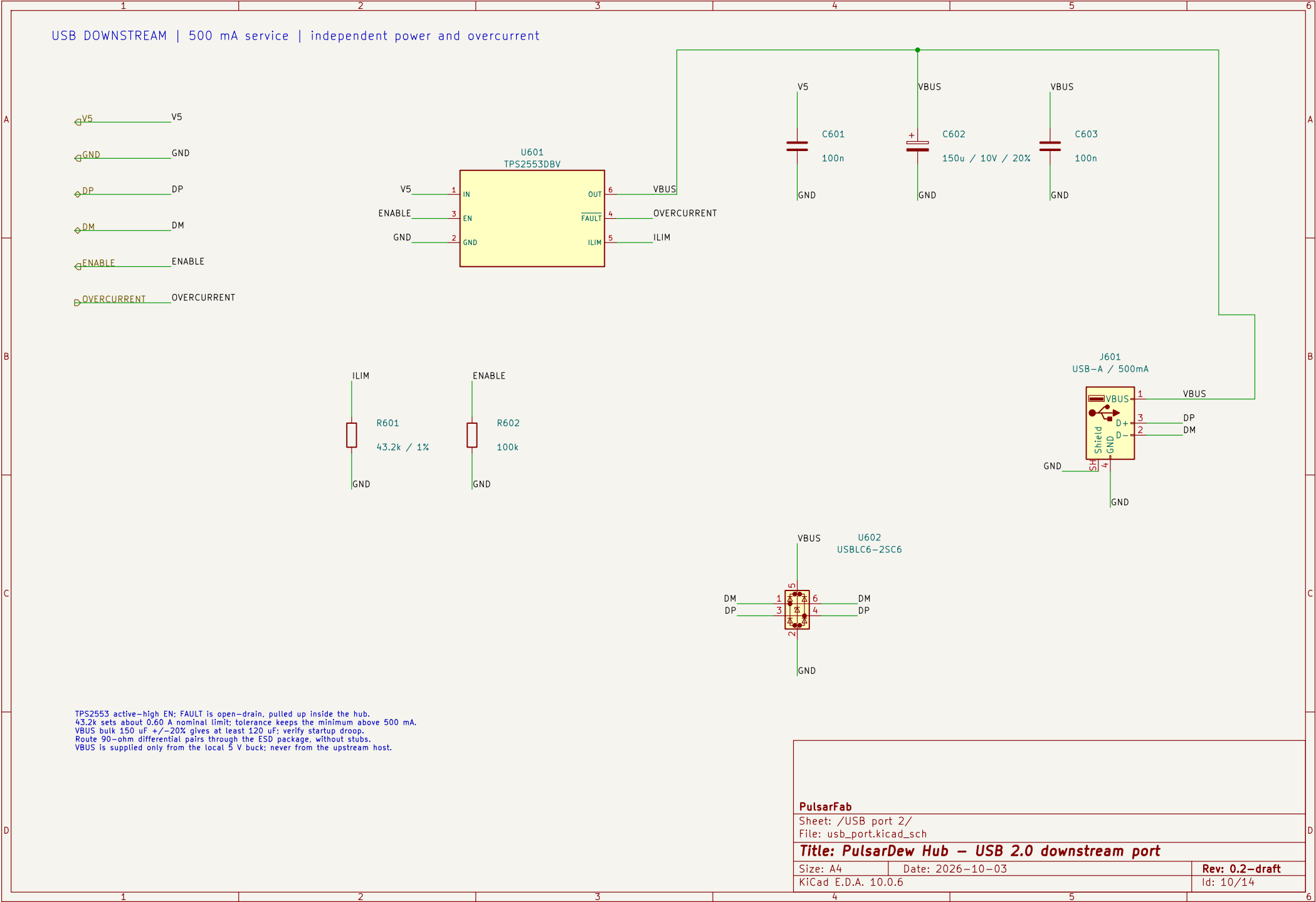
Sheet: /DC outlet 2/
File: dc_outlet.kicad_sch

Title: PulsarDew Hub – 5 A switched DC outlet

Size: A3	Date: 2026–10–03	Rev: 0.2–draft
KiCad E.D.A. 10.0.6		Id: 9/14

PulsarFab
 Sheet: /DC outlet 2/
 File: dc_outlet.kicad_sch
Title: PulsarDate Hub – 5 A switched DC outlet
 Size: A3 Date: 2026-10-03 Rev: 0.2-draft
 KiCad E.D.A. 10.0.6 Id: 9/14

USB DOWNSTREAM | 500 mA service | independent power and overcurrent



SWITCHED DC OUTLET | 12–18 V / 5 A service | hardware voltage inhibit / latched breaker

Legend:

- VIN: VIN
- V3: V3
- GND: GND
- ENABLE: ENABLE
- CURRENT: CURRENT
- PGOOD: PGOOD

Components:

- U301: TPS2598270NRGE
- U302: TPS3700DDC
- R301: 220k
- R302: 10k
- R303: 470k
- R304: 10k
- R305: 100k
- R306: 10k
- R307: 249 / 1%
- R308: 820 / 1%
- R309: 10k
- R310: 10k
- C301: 100n / 50V
- C302: 100n / 50V
- C303: 4.7n / 50V
- C304: 10n
- C305: 100n
- D301: S554
- D302: SMBJ18A
- J301: 5A DC / center +

Notes:

- TPS2598270: 249R gives 5.97 A nominal breaker threshold; 5 A service target needs tolerance/bench qualification.
- ITIMER open: fastest overload response. Fast short trip ~2.1x setpoint. RETRY_DLY grounded: latch off.
- IMON: 246 uA/A x 820R = 0.20172 V/A (1.009 V at 5 A); 10k / 10n ADC filter. Calibrate each outlet.
- TPS3700 independently inhibits EN below ~9.2 V or above ~19.2 V; window recovery can re-enable an asserted GPIO.
- PGOOD is HIGH when ready; LOW also means OFF/startup/inhibit, not exclusively a latched current fault.
- EP25 = VIN, EP26 = GND. -15 C minimum junction rating accepted. No reverse blocking; do not backfeed OUT.
- SMBJ18A input TVS + S554 output clamp need short loops and hot-plug/inductive-load bench validation.

Revision History:

Rev	Description
0.1	Initial design
0.2	Final design

Project Information:

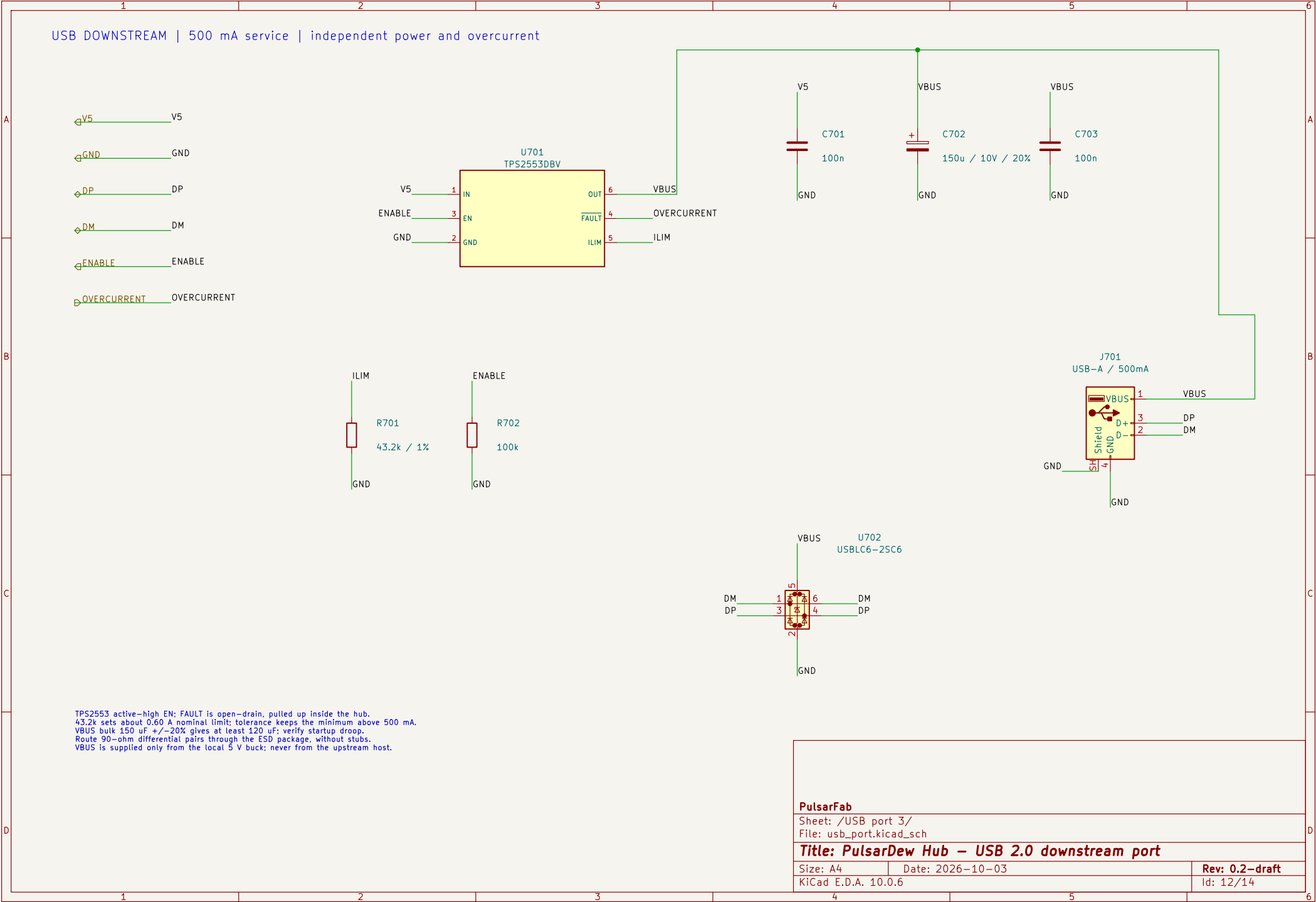
Title: PulsarDew Hub – 5 A switched DC outlet

Size: A3 | Date: 2026–10–03 | Rev: 0.2–draft

KiCad E.D.A. 10.0.6 | Id: 11/14

TPS2598270: 249R gives 5.9/ A nominal breaker threshold; 5 A service target needs tolerance/bench qualification. ITIMER open: fastest overload response. Fast short trip ~2.1x setpoint. RETRY_DLY grounded: latch off. IMON: 246 uA x 820R = 0.20172 V/A (1.009 v at 5 A); 10k / 10n ADC divider. Calibrate each outlet. TPS37700 independently inhibits EN below -9.2 V or above -19.2 V; window recovery can re-enable an asserted GPIO. PGOOD is HIGH when ready; LOW also means OFF/startup/inhibit, not exclusively a latched current fault. EPB2 = VIN, EP26 = GND, -15 C minimum junction rating accepted. No reverse blocking; do not backfeed OUT. SMBJ18A input TVS + S554 output clamp need short loops and hot+plug/inductive-load bench validation.

USB DOWNSTREAM | 500 mA service | independent power and overcurrent



PulsarFab

Sheet: /USB port 3/
File: usb_port.kicad_sch

Title: PulsarDew Hub – USB 2.0 downstream port

Size: A4

Date: 2026-10-03

Rev: 0.2-draft

KiCad E.D.A. 10.0.6

Id: 12/14

SWITCHED DC OUTLET | 12-18 V / 5 A service | hardware voltage inhibit / latched breaker

VIN

V3

GND

ENABLE

CURRENT

PGOOD

VIN

D402

SMBJ18A

GND

VIN

C401

100n / 50V

GND

U401

TPS2598270NRGE

VIN

1

IN

2

IN

3

IN

16

IN

25

EP_IN

6

EN_SW

4

GND

5

GND

14

GND

26

EP_GND

17

OUT

18

OUT

19

OUT

20

OUT

21

OUT

22

OUT

23

OUT

24

OUT

13

PGOOD

9

IMON_RAW

8

ILIM

15

DVDT

7

ITIMER

10

RETRY_DLY

11

NRETRY

12

LDSTRT

OUT

C402

100n / 50V

GND

J401

5A DC / center +

1

OUT

3

GND

2

GND

V3

R406

10k

PGOOD

OUT

D401

S554

GND

VIN

R401

220k

UV_SENSE

VIN

R403

470k

OV_SENSE

V3

5

VDD

3

INA+

4

INB-

2

GND

U402

TPS3700DDC

1

OUTA

6

OUTB

EN_SW

EN_SW

UV_SENSE

OV_SENSE

GND

R402

10k

GND

OV_SENSE

R404

10k

GND

V3

C405

100n

GND

ENABLE

R410

10k

EN_SW

DVDT

C403

4.7n / 50V

GND

ILIM

R407

249 / 1%

GND

EN_SW

R405

100k

GND

IMON_RAW

R409

10k

CURRENT

IMON_RAW

R408

820 / 1%

GND

CURRENT

C404

10n

GND

TPS2598270: 249R gives 5.97 A nominal breaker threshold; 5 A service target needs tolerance/bench qualification.
 ITIMER open: fastest overload response. Fast short trip ~2.1x setpoint. RETRY_DLY grounded: latch off.
 IMON: 246 uA/A x 820R = 0.20172 V/A (1.009 V at 5 A); 10k / 10n ADC filter. Calibrate each outlet.
 TPS3700 independently inhibits EN below ~9.2 V or above ~19.2 V; window recovery can re-enable an asserted GPIO.
 PGOOD is HIGH when ready; LOW also means OFF/startup/inhibit, not exclusively a latched current fault.
 EP25 = VIN, EP26 = GND. -15 C minimum junction rating accepted. No reverse blocking; do not backfeed OUT.
 SMBJ18A input TVS + S554 output clamp need short loops and hot-plug/inductive-load bench validation.

PulsarFab

Sheet: /DC outlet 4/
 File: dc_outlet.kicad_sch

Title: PulsarDew Hub - 5 A switched DC outlet

Size: A3 Date: 2026-10-03 Rev: 0.2-draft

KiCad E.D.A. 10.0.6 Id: 13/14

PulsarFab
 Sheet: /DC outlet 4/
 File: dc_outlet.kicad_sch
Title: PulsarDev Hub – 5 A switched DC outlet
 Size: A3 Date: 2026-10-03 Rev: 0.2-draft
 KiCad E.D.A. 10.0.6 Id: 13/14

USB DOWNSTREAM | 500 mA service | independent power and overcurrent

